

Capital Allocation. Optimal Capital Allocation using Maixum Utility

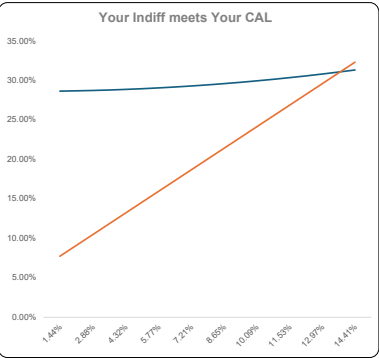
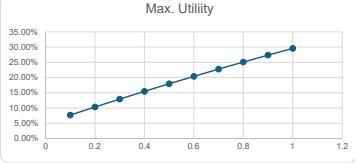
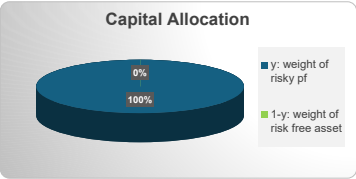
Capital Allocation > Asset Classes Allocation > Security Selection

Risky Pf (P) only	
rf: rate of return of risk-free asset	3.69%
E(Rp): expected return of risky pf	31.29%
	14.41%

Complete Pf (C), Risk free + Risky Pf	
y: weight of risky pf	100.00%
1-y: weight of risk free asset	0.00%
E(rc): expected return of the Complete pf	31.29%
	14.41%
Slope: Capital Allocation Line (Sharpe)	1.9148
Risk Tolerance Score	87
Risk Aversion Index (A: 1-20)*	2.6
Utility at the risk aversion level	28.59%

*Bear (1-5), Normal (1-10), Bull (1-20), Bubble (1-30)

W of risky pf: y	E(rc) on CAL	Utility	28.59%
			E(rc) on Indiff.
Max. U	31.29%	14.41%	28.59%
0.1	7.73%	1.44%	7.70%
0.2	10.46%	2.88%	10.35%
0.3	13.18%	4.32%	12.94%
0.4	15.91%	5.77%	15.48%
0.5	18.64%	7.21%	17.97%
0.6	21.37%	8.65%	20.40%
0.7	24.10%	10.09%	22.77%
0.8	26.83%	11.53%	25.10%
0.9	29.55%	12.97%	27.37%
1.0	32.28%	14.41%	29.58%



RISK TOLERANCE QUESTIONNAIRE

The Risk Tolerance Questionnaire is designed to help you assess your client's risk tolerance and investment objective. These questions are assigned numerical weights to reflect each one's comparative importance in overall risk determination. These weights are identified in parentheses next to each answer choice. Completing the questionnaire in its entirety will allow you to better evaluate your client's profile.

Client Name: **sai vrishank**

Registered Representative: _____ Number: _____

Statement of Investment Selection (SIS) Number (Optional): _____

1. For these funds, which of the following closely aligns with your current financial goal? *Please select one.*
- ☐ Sustaining current income and account preservation (0)
 - ☐ Sustaining current income with possible growth opportunity (10)
 - ☐ Growing account value, not tied to current income needs (20)
 - ☒ Aggressive growth, maximizing accumulation (30)
2. How long do you plan to keep these funds invested in order to achieve your financial goal?
- ☐ Less than 1 year (0)
 - ☐ 1 to 2 years (3)
 - ☐ 3 to 5 years (8)
 - ☐ 6 to 10 years (15)
 - ☒ 11 to 20 years (23)
 - ☐ Greater than 20 years (30)

3. Every investment has an opportunity for both risk and reward. The chart below represents a one-year hypothetical risk and reward scenario for five portfolios with incremental levels of risk and reward for a hypothetical initial investment of \$100,000. Select the option with which you are most comfortable.

Note: these numbers are not representative of your potential target portfolios. *Please select one.*



4. How would you react to a significant fall in the value of the stock market?



If your hypothetical investment of \$100,000 experienced a sudden and unexpected drop of 24% over a three-month period, what would your reaction be?

- ☐ Sell All, Avoid Further Risk (0)
- ☐ Sell Some, Reduce Exposure to Risk (3)
- ☒ Sell Nothing, Remain Invested (7)
- ☐ Buy More, Opportunity is Present (10)

5. How soon would you need these funds to recover after experiencing a sudden meaningful loss in value?

- ☐ 0 to 6 months (0)
- ☐ 6 months to 1 year (3)
- ☐ 1 to 3 years (7)
- ☒ 3 years or more (10)

6. How would you respond to the following statement: I am comfortable investing during times of uncertainty.

- ☐ Strongly disagree (0)
- ☐ Disagree (3)
- ☒ Agree (7)
- ☐ Strongly agree (10)

YOUR SCORE

87

POINT SCALE

0 THROUGH 20
21 THROUGH 40
41 THROUGH 60
61 THROUGH 80
81 THROUGH 100

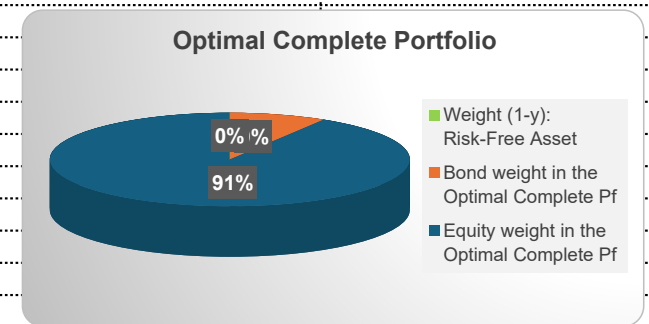
RISK TOLERANCE

CONSERVATIVE
MODERATELY CONSERVATIVE
MODERATE
MODERATELY AGGRESSIVE
AGGRESSIVE

Investing involves risk including the potential loss of principal. No investment strategy can guarantee a profit or protect against loss. Risk and reward figures are for illustrative purposes only and are not indicative of any specific investment product or portfolio.

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	B	C	D	E	F	G
2	Asset Classes Allocation Analysis: Optimal Complete Pf Model					
3	Capital Allocation > Asset Classes Allocation > Security Selection					
4						
5	Securities	Expected Return	Standard Deviation	Corr. Coefficient B,E	Covariance B,E	
6	Bond Pf (B)	14.10%	13.73%	0.3224	0.0068	
7	Equity Pf (E)	32.94%	15.32%			
8	T-Bill	3.69%	0.00%			
9						
10	Risky Asset Portfolio: Weights				Reward to Variability	
11	Bond weight	Equity weight	Expected Return	Standard Deviation	(Sharpe ratio)	
12	1.0	0.0	0.1410	0.1373	0.7585	
13	0.9	0.1	0.1599	0.1293	0.9510	
14	0.8	0.2	0.1787	0.1232	1.1513	
15	0.7	0.3	0.1975	0.1191	1.3483	
16	0.6	0.4	0.2164	0.1175	1.5280	
17	0.5	0.5	0.2352	0.1182	1.6777	
18	0.4	0.6	0.2540	0.1213	1.7896	
19	0.3	0.7	0.2729	0.1267	1.8627	
20	0.2	0.8	0.2917	0.1340	1.9019	
21	0.1	0.9	0.3106	0.1429	1.9146	
22	0.0	1.0	0.3294	0.1532	1.9088	
23	A. Minimum Variance Portfolio of Risky Assets		Expected Return	Min. Standard Deviation	Sharpe Ratio	
24	0.58	0.42	0.2200	0.1174	1.5596	sho
25	B. Optimal Risky Portfolio. Maximum of Sharpe ratio		Expected Return	Standard Deviation	Max. of Sharpe Ratio	
26	0.09	0.91	0.3129	0.1441	1.9148	sho
27						
28	C. Optimal Complete Portfolio with U w/ Risk-Free Asset					
29	Weight y: Optimal Risky Portfolio		100.00%			
30	Weight (1-y): Risk-Free Asset		0.00%			
31	Bond weight in the Optimal Complete Pf		8.75%			
32	Equity weight in the Optimal Complete Pf		91.25%			
33	E(rc): expected return of the Overall Complete pf		0.3129			
34			0.1441			
35	Sharpe Ratio (Max. Slope of CAL)		1.9148			
36						
37						
38						
39						
40						



Index Portfolio Model. SPDR + Active Portfolio (Large Growth)

Panel A. Risk Parameters of Pf A (annualized)	IWF	APH	LLY	VST	TRGP	PWR	MCK	EME	COR
Beta	1.0000	1.0612	0.3851	1.1500	0.3284	0.8178	0.2004	0.8040	0.3786
SD of Excess Return	0.1940	0.2649	0.3178	0.4407	0.3039	0.3133	0.2425	0.3104	0.2277
SD of Residual (e), Firm specific Component	0.0000	0.1667	0.3089	0.3800	0.2971	0.2701	0.2393	0.2683	0.2156
SD of Systematic Component	0.1940	0.2059	0.0747	0.2231	0.0637	0.1587	0.0389	0.1560	0.0735
Correlation with the Market Index	1.0000	0.7773	0.2351	0.5064	0.2097	0.5065	0.1604	0.5026	0.3226

Panel B. Correlation Matrix	IWF	APH	LLY	VST	TRGP	PWR	MCK	EME	COR
IWF	1.0000								
APH	0.7773	1.0000							
LLY	0.2351	0.1827	1.0000						
VST	0.5064	0.3936	0.1191	1.0000					
TRGP	0.2097	0.1630	0.0493	0.1062	1.0000				
PWR	0.5065	0.3937	0.1191	0.2565	0.1062	1.0000			
MCK	0.1604	0.1246	0.0377	0.0812	0.0336	0.0812	1.0000		
EME	0.5026	0.3907	0.1182	0.2545	0.1054	0.2546	0.0806	1.0000	0.0000
COR	0.3226	0.2507	0.0758	0.1634	0.0676	0.1634	0.0517	0.1621	1.0000

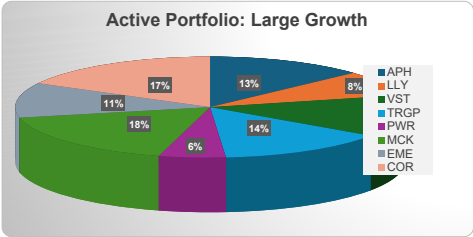
Panel C. The Index Model Covariance Matrix	IWF	APH	LLY	VST	TRGP	PWR	MCK	EME	COR
IWF	0.0377								
APH	0.0400	0.0702							
LLY	0.0145	0.0154	0.1010						
VST	0.0433	0.0460	0.0167	0.1942					
TRGP	0.0124	0.0131	0.0048	0.0142	0.0923				
PWR	0.0308	0.0327	0.0119	0.0354	0.0101	0.0982			
MCK	0.0075	0.0080	0.0029	0.0087	0.0025	0.0062	0.0588		
EME	0.0303	0.0321	0.0117	0.0348	0.0099	0.0248	0.0061	0.0963	
COR	0.0143	0.0151	0.0055	0.0164	0.0047	0.0117	0.0029	0.0115	0.0519

Panel D. Correlation of Residuals in Pf A	IWF	APH	LLY	VST	TRGP	PWR	MCK	EME	COR
IWF	0.5026	0.3907	0.1182	0.2545	0.1054	0.254584173	0.080603009	1	0.0000
APH		1.0000							
LLY		(0.0155)	1.0000						
VST		0.2395	(0.0045)	1.0000					
TRGP		(0.1295)	0.0937	0.2529	1.0000				
PWR		0.3426	(0.0880)	0.3191	0.2035	1.0000			
MCK		0.2957	0.2426	0.1626	0.2983	0.2121	1.0000		
EME		0.3218	(0.1380)	0.4841	0.1787	0.4826	(0.0153)	1.0000	
COR	-	0.3623	0.2668	0.0951	0.2825	0.1199	0.7060	(0.0006)	1.0000

Panel E. Macro Forecast and Alpha Forecasts	IWF	APH	LLY	VST	TRGP	PWR	MCK	EME	COR
Adjusted Alpha	0.0000	0.0910	0.1912	0.4791	0.3098	0.1006	0.2480	0.2001	0.1982
Beta	1.0000	1.0612	0.3851	1.1500	0.3284	0.8178	0.2004	0.8040	0.3786
Total Equity Risk Premium E(R)	0.0863	0.1825	0.2244	0.5783	0.3381	0.1711	0.2653	0.2695	0.2309
Adjusted Market risk premium =	0.0863								

Panel F. Active Portfolio			Active Pf	APH	LLY	VST	TRGP	PWR	MCK	EME	COR
Residual Variance of each security	$\sigma^2(e)$			0.0278	0.0954	0.1444	0.0883	0.0730	0.0573	0.0720	0.0465
Initial position of each security	$w_i(0) = \alpha_i / \sigma^2(e_i)$		24.8579	3.2736	2.0034	3.3180	3.5091	1.3780	4.3299	2.7799	4.2660
Scaled initial position of each security	w_i		1.0000	0.1317	0.0806	0.1335	0.1412	0.0554	0.1742	0.1118	0.1716
Squared scaled initial position of each security	$[w_i(0)]^2$			0.0173	0.0065	0.0178	0.0199	0.0031	0.0303	0.0125	0.0295
Alpha of the Active Pf	α_A		0.2402								
Residual Variance of Active Pf	$\sigma^2(e_A)$		0.0097								
Active Pf information ratio (APIR)	$\alpha_A / \sigma^2(e_A)$		24.8579								
Index Pf information ratio (IPIR)	$E(R_M) / s^2(M)$		2.2912								
Initial position in Active Pf (APIR/IPIR)	$w_A(0)$		10.8492								
Beta of the Active Pf	β_A		0.6058								

Panel G. Optimal Risky Portfolios	Overall Risky Pf	IWF	Active Pf	APH	LLY	VST	TRGP	PWR	MCK	EME	COR
w*(Optimal Pf): Short not allowed		0.0000	1.0000	0.1317	0.0806	0.1335	0.1412	0.0554	0.1742	0.1118	0.1716
Beta	0.6058	1.0000	0.6058								
Risk Premium	0.2925	0.0863	0.2925								
Standard Deviation	0.1532	0.1940	0.1532								
Sharpe Ratio (S)	1.9088	0.4446	1.9088								
Performance Measures											
Treynor Measure (T)	0.4829	0.0863	0.4829								
Jensen Measure (a)	0.2402	0.0000	0.2402								
Information Ratio Measure (IR)	1.3458	0.0000	1.3458								
Overall Performance	0.2925	0.0863	0.2925								
CAPM	0.0892	0.1232	0.0892								
	0.7897	1.0000	0.7897								
CML	0.1050	0.1232	0.1050								
Selectivity (%)	0.2402	0.0000	0.2402								
Diversification Effect (%)	-0.0159	0.0000	-0.0159								
Net Selectivity (%)	0.2561	0.0000	0.2561								
net Risk Premium of Pf	0.0523	0.0863	0.0523								



Panel 1. Risk & Return Parameters in the Security Index Model

$$\text{Cov}(x_i, x_j) = \text{Cov}(\beta_i w + e_i, \beta_j w + e_j) = \beta_i \beta_j \sigma_w^2$$

Panel 2. Correlation and Covariance Matrix in the Security Index Model

[illegible]

Panel 3. Price information, monthly rate of return and excess return

[illegible]

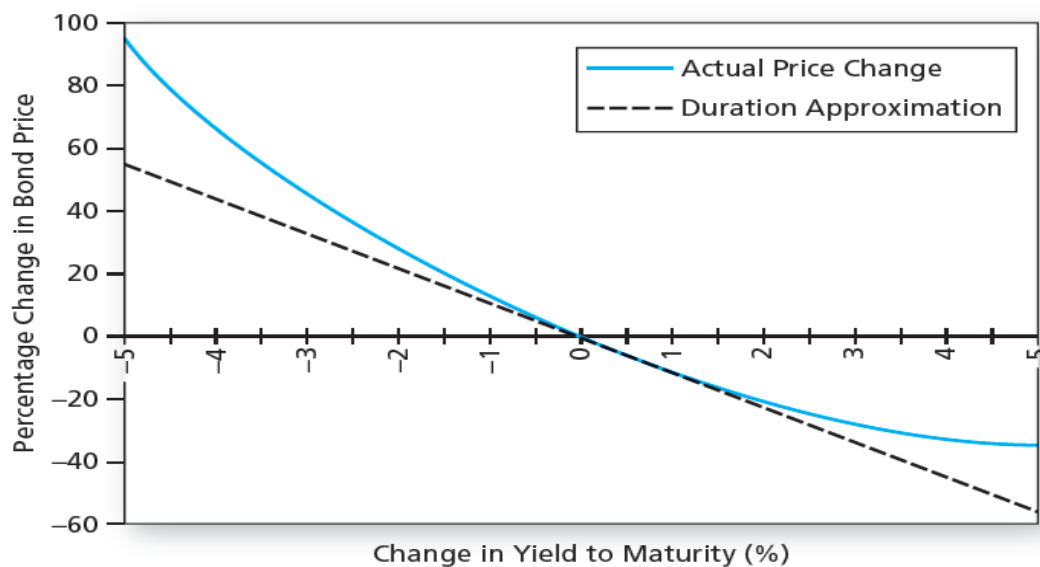
Calculate the approximate price change for a VGLT using only its D* and using D* and Convexity

VGLT has the following characteristics:

Maturity	22
Coupon (\$100 par)	4.00%
Yield to maturity	4.70%
Coupon (frequency)	Semiannual
Convexity (C)	3

Current Bond Price	\$90.47	=PV(C7/2,C5*2,-C6/2*100,-100)
Modified Duration (D*)	14.08652511	=MDURATION(DATE(2025,4,1),DATE(2047,4,1),C6,C7,2)

Changes in YTM	YTM down 100bps	
with D* only	14.0865%	=-C11*-1%
Bond Price change	103.2095	=C10*(1+C14)
with D* and Convexity	14.1015%	=C14+1/2*C9*-1%^2
Bond Price change	103.2231	=C10*(1+C16)



ADDITIONAL ANALYSIS — SUPPORTING DATA FOR INVESTMENT POLICY STATEMENT

1. RISK TOLERANCE QUESTIONNAIRE — SCORING BREAKDOWN

#	Question	Client Response	Weight
1	Current financial goal	Aggressive growth, maximizing accumulation	30
2	Investment time horizon	11 to 20 years	23
3	Risk/return preference	Portfolio #5 (highest risk/return)	10
4	Reaction to market decline	Sell Nothing, Remain Invested	7
5	Recovery period tolerance	3 years or more	10
6	Comfort with uncertainty	Agree	7
TOTAL RISK TOLERANCE SCORE			87

Classification (based on score):AGGRESSIVE

Risk Aversion Index Calculation (Bull Market Scale)

Risk Tolerance Score	87	Scale: Bear=5, Normal=10, Bull=20, Bubble=30
Bull Market Scale Cap	20	
Formula: A = Cap × (1 – Score/100)	2.6	

2. CAPITAL ALLOCATION SENSITIVITY — γ* AT DIFFERENT RISK AVERSIONS

Inputs from Capital Allocation sheet:	
Expected Return of Risky Pf, E[Rp]	31.29%
Risk-free Rate, Rf	3.69%
Standard Deviation, σp	14.41%
Excess Return [E[Rp] – Rf]	27.60%
Variance (σp²)	0.0208

Risk Aversion Scale Label	A (Index)	Unconstrained γ*	Constrained γ	E(rc)	σc	Utility
Very Conservative	15.0	0.89	88.61%	28.15%	12.77%	15.92%
Conservative	10.0	1.33	100.00%	31.29%	14.41%	20.91%
Moderate	6.0	2.22	100.00%	31.29%	14.41%	25.06%
Moderately Aggressive	4.0	3.32	100.00%	31.29%	14.41%	27.14%
Sai Vrishank (Actual)	2.6	5.11	100.00%	31.29%	14.41%	28.59%
Very Aggressive	1.5	8.86	100.00%	31.29%	14.41%	29.73%

3. ALPHA SHRINKAGE SENSITIVITY (Grinold 1989)

Forward-realized alpha is typically a fraction of backward-looking alpha due to mean-reversion.

Scenario	Shrinkage Factor	Adj. Portfolio Alpha	Adj. Expected Return	Adj. Sharpe Ratio	Forward IR
In-sample (no shrinkage)	100%	24.02%	31.29%	1.92	1.35
Mild shrinkage (Grinold upper)	50%	12.01%	19.28%	1.08	0.67
Moderate shrinkage (Grinold base)	33%	7.93%	15.20%	0.80	0.44
Aggressive shrinkage (Grinold lower)	25%	6.01%	13.28%	0.67	0.34
Zero alpha (efficient market)	0%	0.00%	7.27%	0.25	0.00

Source: Grinold, R. C. (1989). The Fundamental Law of Active Management. Journal of Portfolio Management, 15(3), 30–37.

4. SECURITY DEEP DIVE — 8 PORTFOLIO HOLDINGS

Ticker	Company	Sector	Industry	Mkt Cap (\$B)	Alpha (α)	Beta (β)	Theme
APH	Amphenol Corp.	Info Tech	Electronic Components	\$85	19.07%	1.06	AI infrastructure
LLY	Eli Lilly & Co.	Healthcare	Pharmaceuticals	\$700	28.60%	0.39	GLP-1 franchise
VST	Vistra Corp.	Utilities	Independent Power	\$65	40.73%	1.15	AI power demand
TRGP	Targa Resources	Energy	Midstream	\$45	34.46%	0.33	LNG export growth
PWR	Quanta Services	Industrials	Construction Engineering	\$48	30.22%	0.82	Grid modernization
MCK	McKesson Corp.	Healthcare	Healthcare Distributors	\$75	27.97%	0.20	Pharma distribution
EME	EMCOR Group	Industrials	Construction Engineering	\$24	30.27%	0.80	Datacenter construction
COR	Cencora, Inc.	Healthcare	Healthcare Distributors	\$50	16.29%	0.38	Specialty pharma

Sector Concentration Check (Risk Management — no sector > 50%)

Sector	# of Holdings	% of Equity Sleeve
Healthcare	3	37.5%
Industrials	2	25.0%
Info Tech	1	12.5%
Utilities	1	12.5%
Energy	1	12.5%

5. ECONOMIC & INDUSTRY OUTLOOK

Macroeconomic Assumptions				
Indicator	Current	12-Month Outlook	Direction	Portfolio Implication
U.S. GDP Growth (YoY)	2.5%	2.0-2.5%	Stable	Supports equity earnings growth
Core CPI Inflation (YoY)	2.8%	2.3-2.7%	Declining	Supports Fed easing; positive for bonds
Fed Funds Rate	4.25-4.50%	3.50-4.00%	Declining	Bullish for long-duration VGLT sleeve
10-Year Treasury Yield	4.20%	3.80-4.20%	Stable/Down	Duration positioning benefits
Unemployment Rate	4.1%	4.0-4.3%	Stable	Supports consumer spending
S&P 500 Earnings Growth	10%	8-12%	Positive	Supports active growth equity sleeve

Sector	Holdings	5-Year Theme	Outlook	Risk Factors
Healthcare	LLY, MCK, COR	GLP-1 expansion, aging demographics, specialty drug growth	Positive	Patent cliffs, regulatory pressure on drug pricing
Industrials	PWR, EME	Grid modernization, datacenter construction, U.S. reshoring	Positive	Labor costs, supply chain, interest rate sensitivity
Information Tech	APH	AI datacenter buildout, electronic components demand	Positive	Semiconductor cycle, capex normalization risk
Utilities	VST	AI-driven power demand, electrification, nuclear revival	Positive	Regulatory, commodity prices, weather/demand forecasts
Energy	TRGP	LNG export growth, midstream infrastructure	Positive	Oil price volatility, policy/regulatory changes

6. REBALANCING DECISION MATRIX

Trigger Type	Threshold	Current Level	Breach Status	Action Required
Equity sleeve drift	±5 pp from 91.25%	91.25%	OK	86.25%-96.25% band
Bond sleeve drift	±5 pp from 8.75%	8.75%	OK	3.75%-13.75% band
Single position max	≤ 25% of equity sleeve	17.50%	OK	Rebalance if exceeds
Single sector max	≤ 50% of equity sleeve	37.50%	OK	Currently Healthcare highest
Drawdown level	Review if > 20%	0.00%	OK	From peak
Drawdown critical	Expedited review > 30%	0.00%	OK	From peak

7. PERFORMANCE METRIC TARGETS vs ACTUAL

Metric	Formula / Definition	Target	Portfolio	Benchmark	Status
Sharpe Ratio	(Rp – Rf) / σp	> 1.00	1.9148	0.4446	EXCEEDS
Jensen's Alpha (ann.)	Rp – [Rf + β(Rm – Rf)]	> 5%	24.02%	0.00%	EXCEEDS
Treynor Measure	(Rp – Rf) / β	> 0.15	0.4829	0.0863	EXCEEDS
Information Ratio	Active Return / Tracking Error	> 0.50	1.3458	—	EXCEEDS
Standard Deviation	Annualized σ of returns	< 20%	14.41%	19.40%	MEETS
Beta vs IWF	Cov(Rp, Rm) / Var(Rm)	0.6 – 1.0	0.6058	1.0000	MEETS

8. PORTFOLIO STRESS TEST SCENARIOS

Scenario	Market Move	Rate Move	Expected Portfolio Impact	Action
Base case	0%	0 bps	Steady state	No action
Bull case (risk-on)	+20%	+50 bps	+17% (equity gain, bond loss)	Rebalance if drift > 5pp
Bear case (recession)	-20%	-100 bps	-17% (equity loss partly offset by bond gain)	Maintain allocation; tax-loss harvest
Tech bubble burst	-35%	-50 bps	-30% (high equity concentration)	Formal risk review
AI trade unwind	-25% growth	Flat	-22% (concentrated in growth names)	Expedited IPS review
Stagflation shock	-10%	+200 bps	-18% (equity loss + bond duration loss)	Review bond duration

Sources: Bodie, Kane & Marcus (2021); Markowitz (1952); Sharpe (1964); Treynor & Black (1973); Grinold (1989); Morningstar; Federal Reserve Economic Data (FRED); BlackRock/iShares; Vanguard.